

Introduction to Internet of Things

Lecture: 3 and 4

Top 10



1

AIoT for Intelligent Evolution

2

Edge Computing for Ultra-Powerful Processing Capabilities

3

Blockchain for Enhanced IoT Security

4

5G to Boost the Potential of IoT

5

Hyper-Personalized Ecosystems for Tailored Experiences

6

Smart Homes for a Smarter Lifestyle

7

Industrial IoT to Revolutionize Operations

8

Growing Adoption of Voice-Activated IoT Devices

9

The Rise of Digital Twins

10

IoT-Powered Smart Cities for Urban Optimization

Modern IoT Trends

AI and Machine Learning Integration:

- ▶ IoT devices increasingly use AI/ML for data analysis and automation

Edge and Fog Computing:

- ▶ Data is processed closer to the devices rather than sending everything to the cloud

5G Connectivity:

- ▶ High-speed, low-latency 5G networks enable more reliable IoT communications

Example: Autonomous vehicles, remote surgeries

Industrial IoT (IIoT) and Industry 4.0

Healthcare IoT/Wearables

Smart cities

Sustainability and Green IoT

Blockchain for IoT Security:

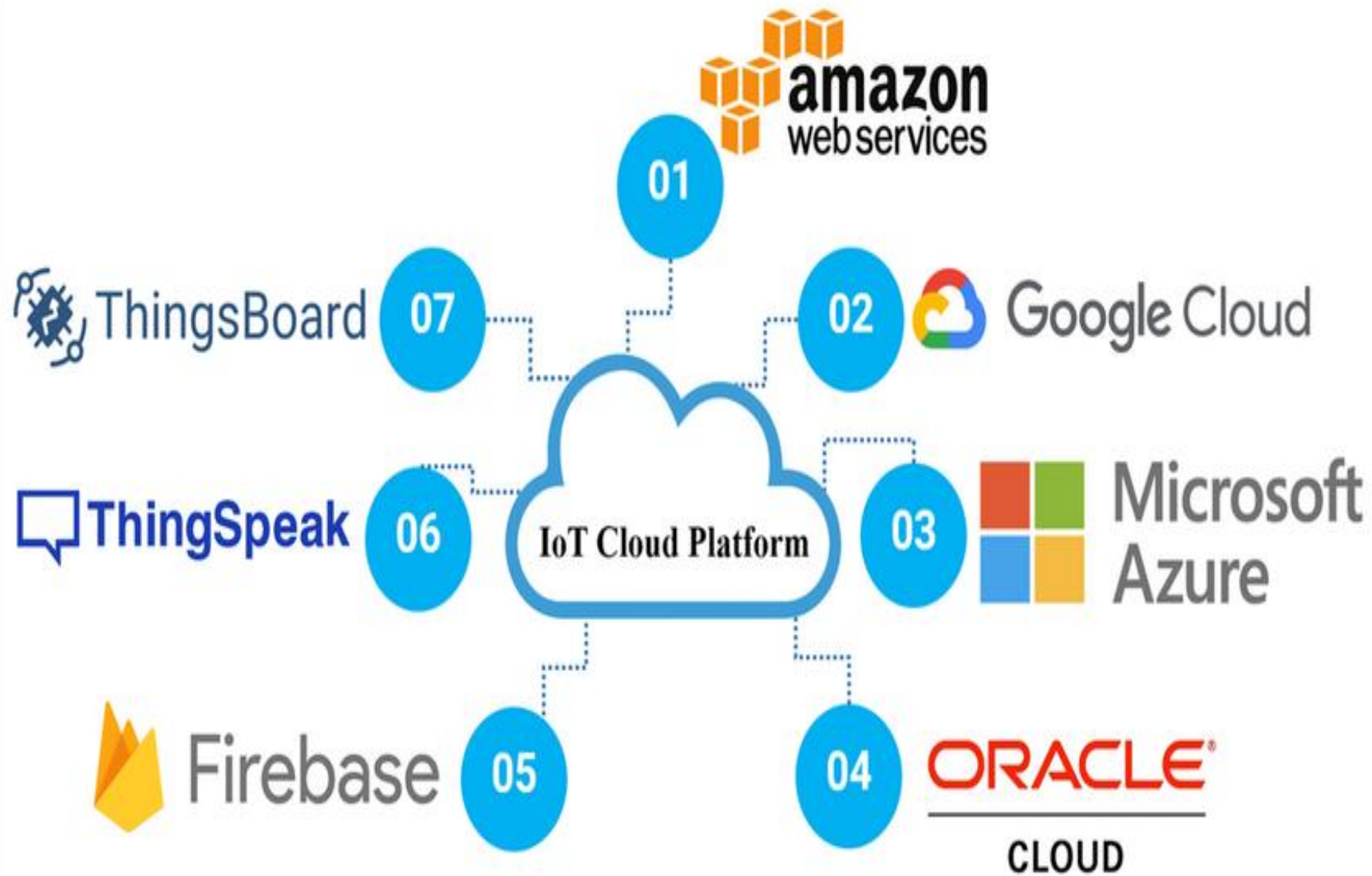
Blockchain is being used to secure IoT devices and data sharing

IoT as a Service (IoTaaS):

- ▶ Cloud-based IoT platforms allow small businesses to deploy IoT without heavy infrastructure

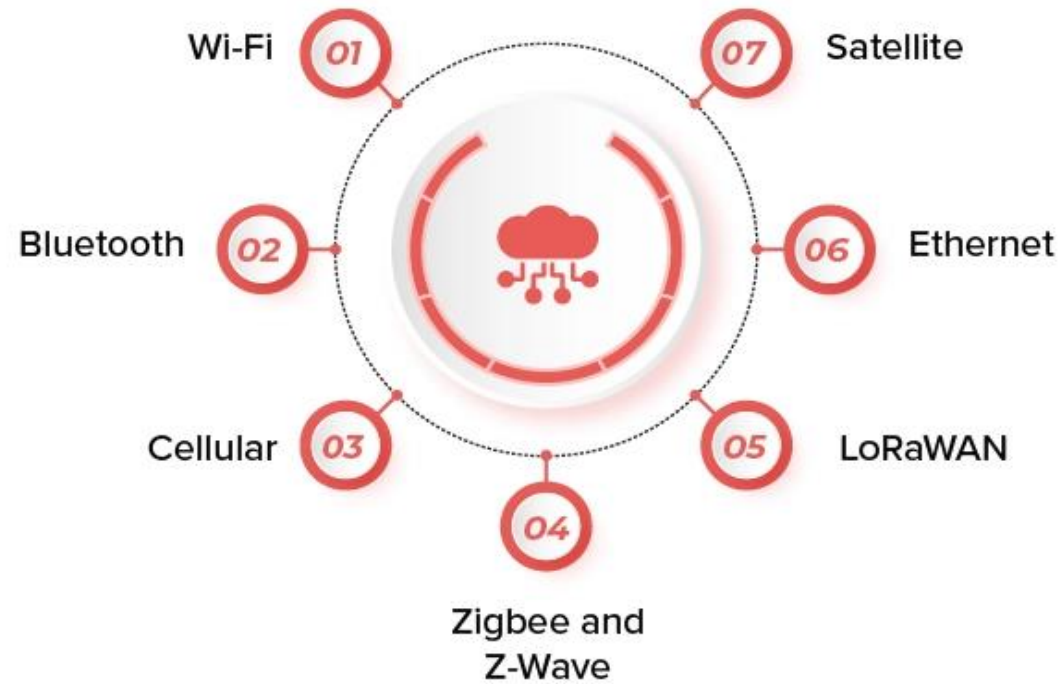
IoT Platforms

- ▶ An IoT platform is a software framework that connects IoT devices, manages the data they generate, and enables applications to interact with these devices.
- ▶ Acts as a bridge between hardware (sensors, actuators) and applications
- ▶ Provides tools for device management, data storage, analytics, and application development

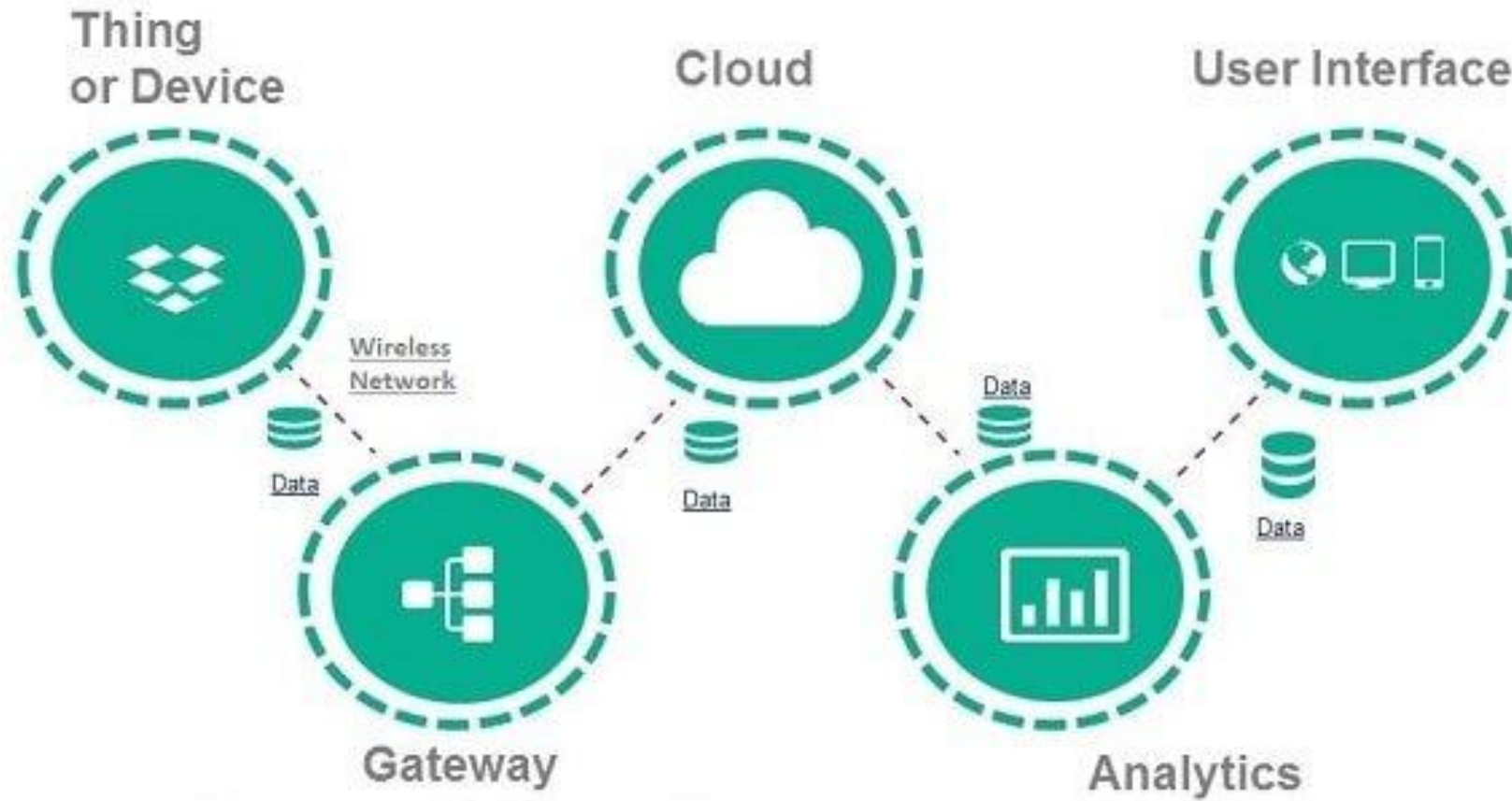


IoT connectivity

Top 7 IoT Connectivity Technologies



Major Components of IoT

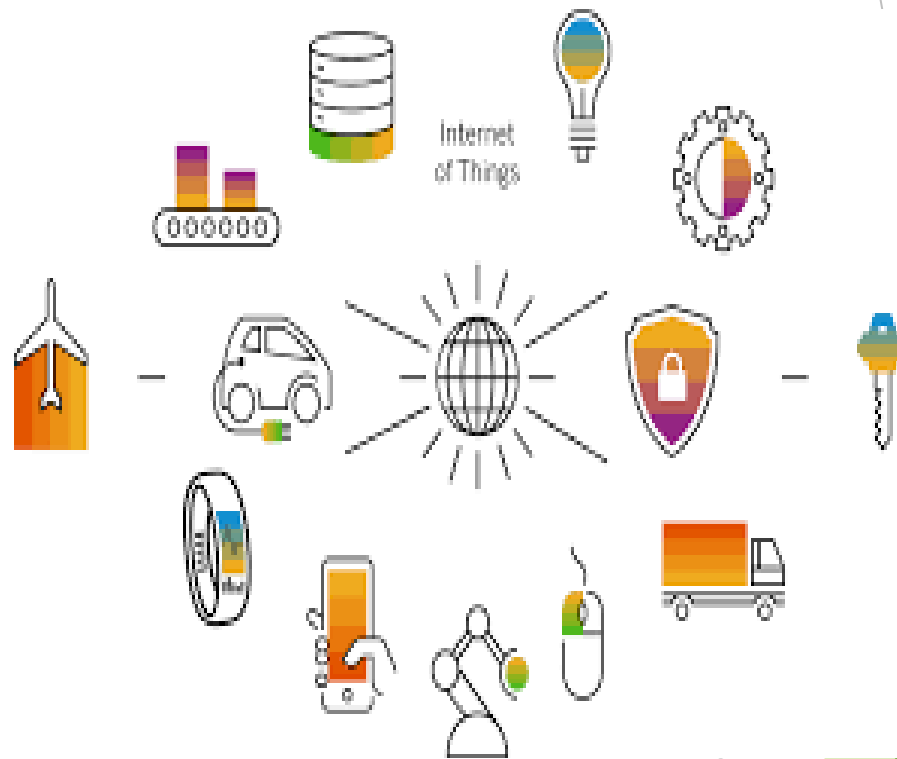


IoT System Architecture



Physical Design of IoT / Hardware Layer

- ▶ **Things** in IoT refers to IoT devices which have unique identities and can perform remote sensing, actuating and monitoring capabilities.
- ▶ IoT devices can exchange data with each other connected devices and applications (directly or indirectly), or collect data from other devices and process the data either locally or send the data to centralized servers or cloud based applications.



Things / Devices in IoT

► Things ----- core of IoT ----- make the system smart and connected

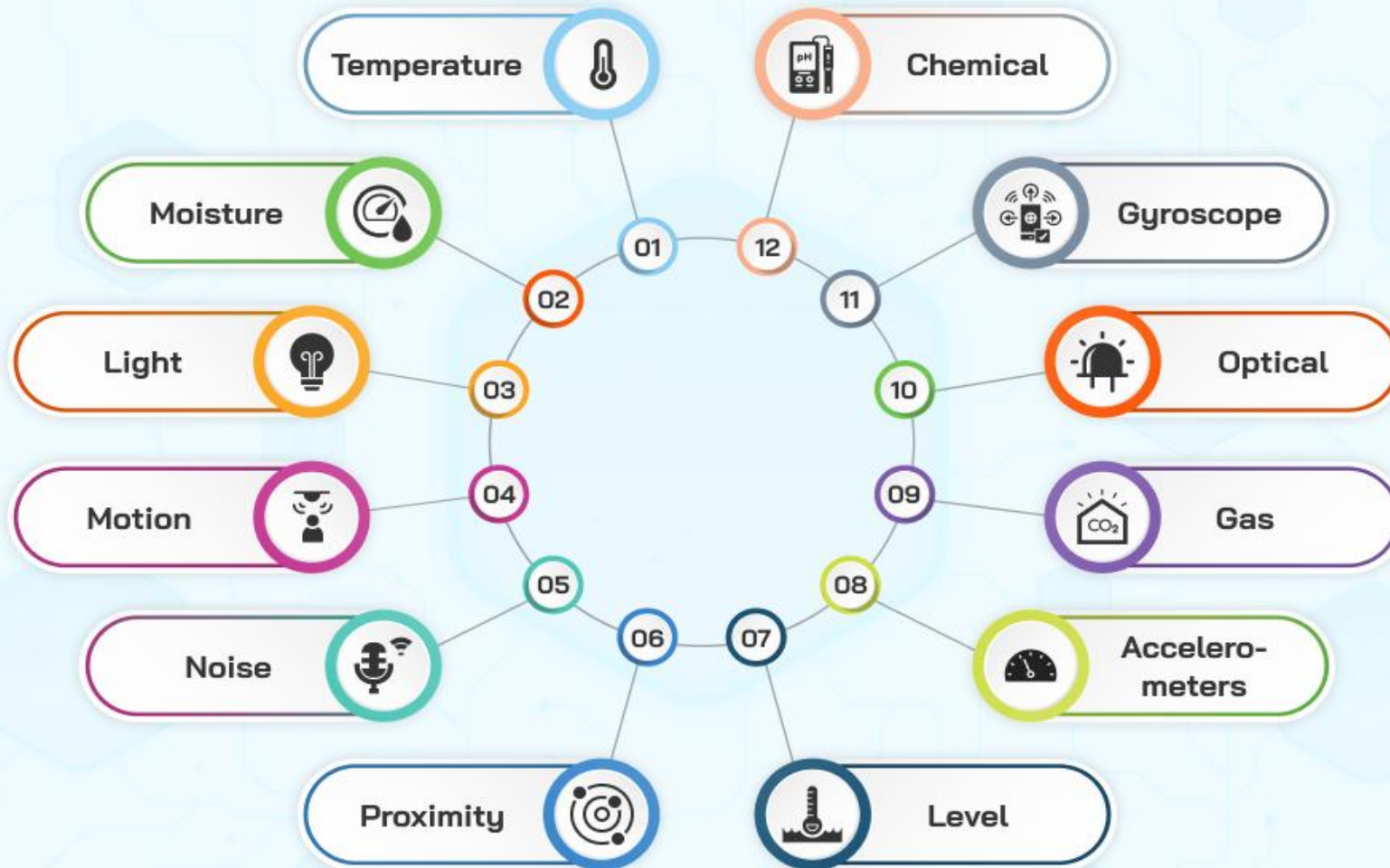
“Thing” usually has:

1. Sensors
2. Actuators
3. Connectivity
4. Processing unit

Characteristics of Things

1. Identifiable
2. Connected
3. Sensing
4. Autonomous / Intelligent

Types of Sensors



Types of Actuators

intuz



Interesting, useful energy



SENSOR



Electric data signal

Electric data signal

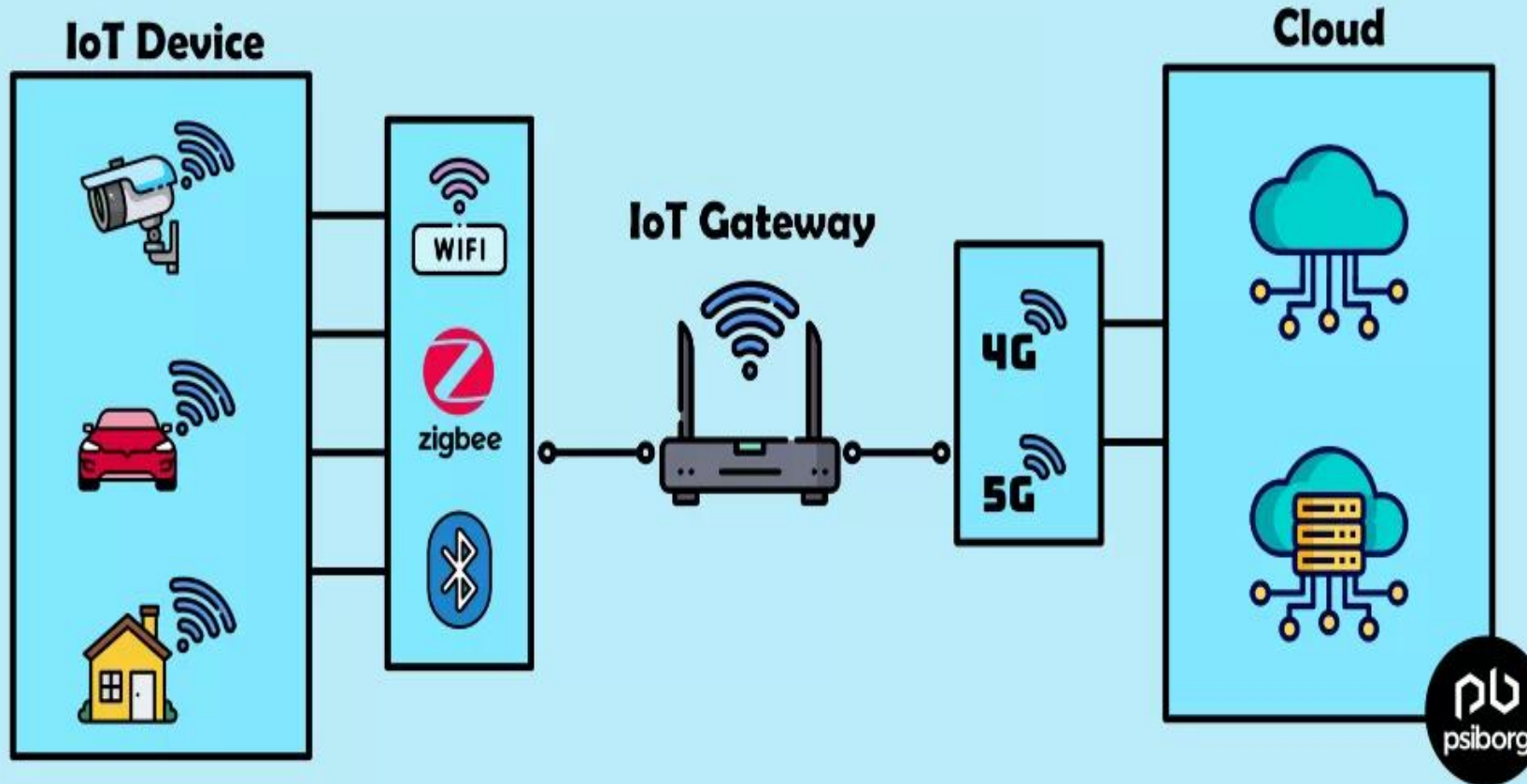


ACTUATOR



Interesting, useful energy

How Does an IoT Gateway Work ?



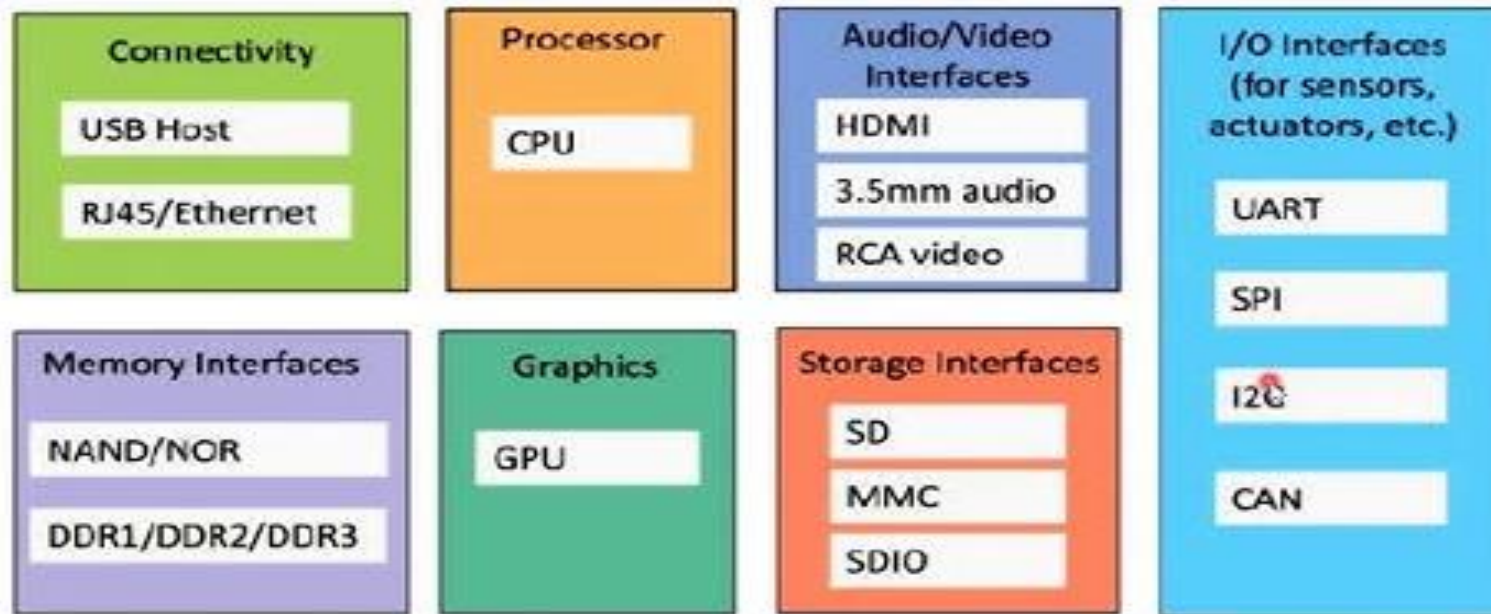
IoT Gateway

Main functions of IoT Gateway are:

1. Data aggregation
2. Protocol translation
3. Edge processing
4. Security
5. Connectivity

Physical Design includes Things/Devices and protocols used to build an IoT system.

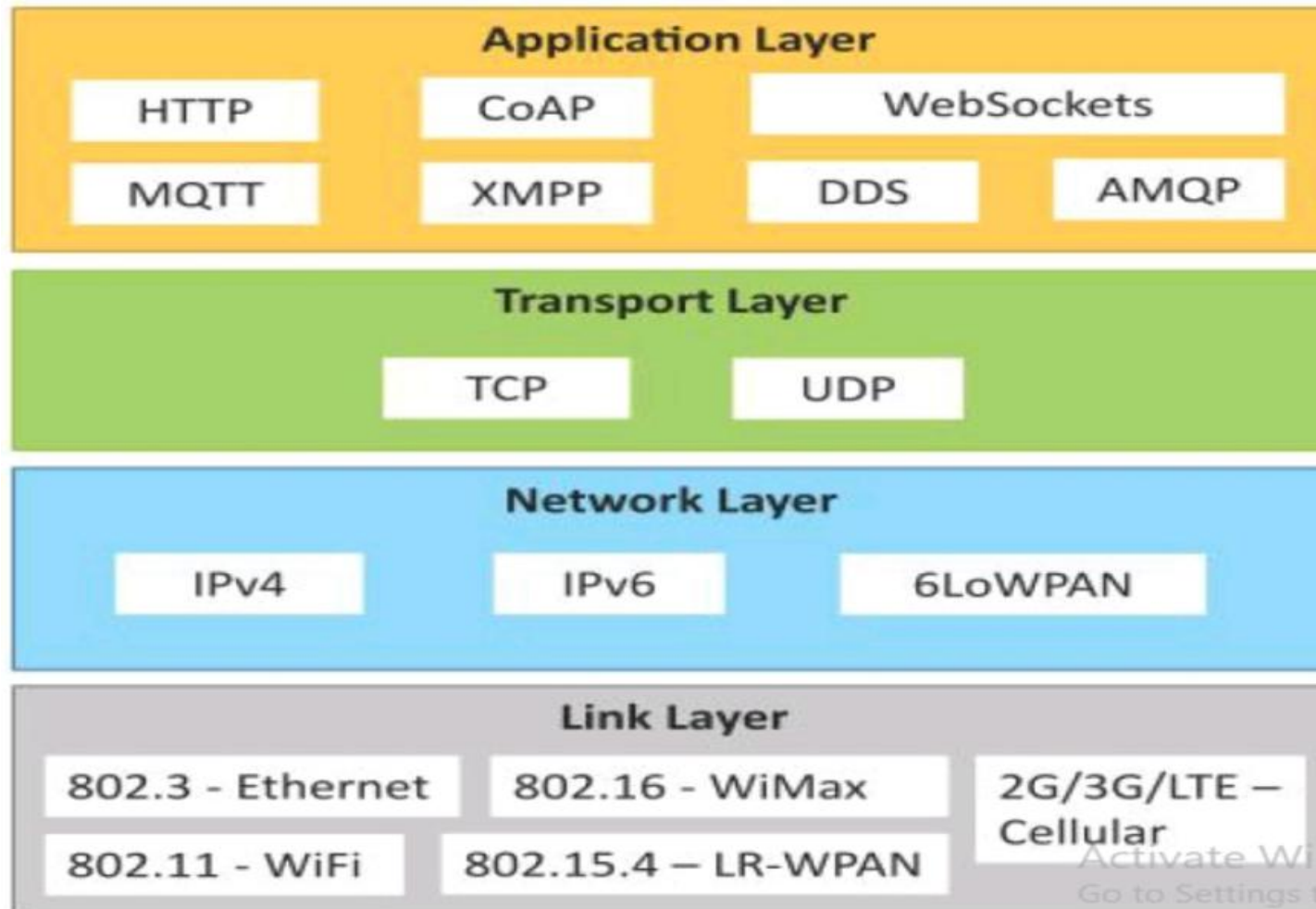
Generic Block Diagram of IoT Device



IoT Devices

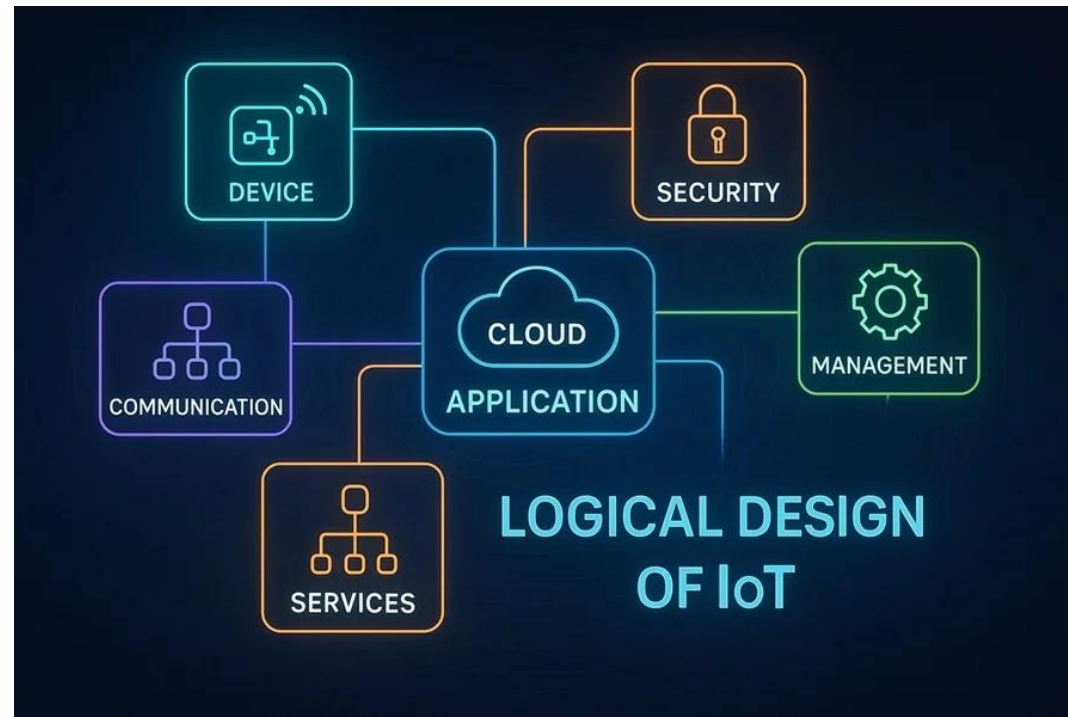


IoT Protocols

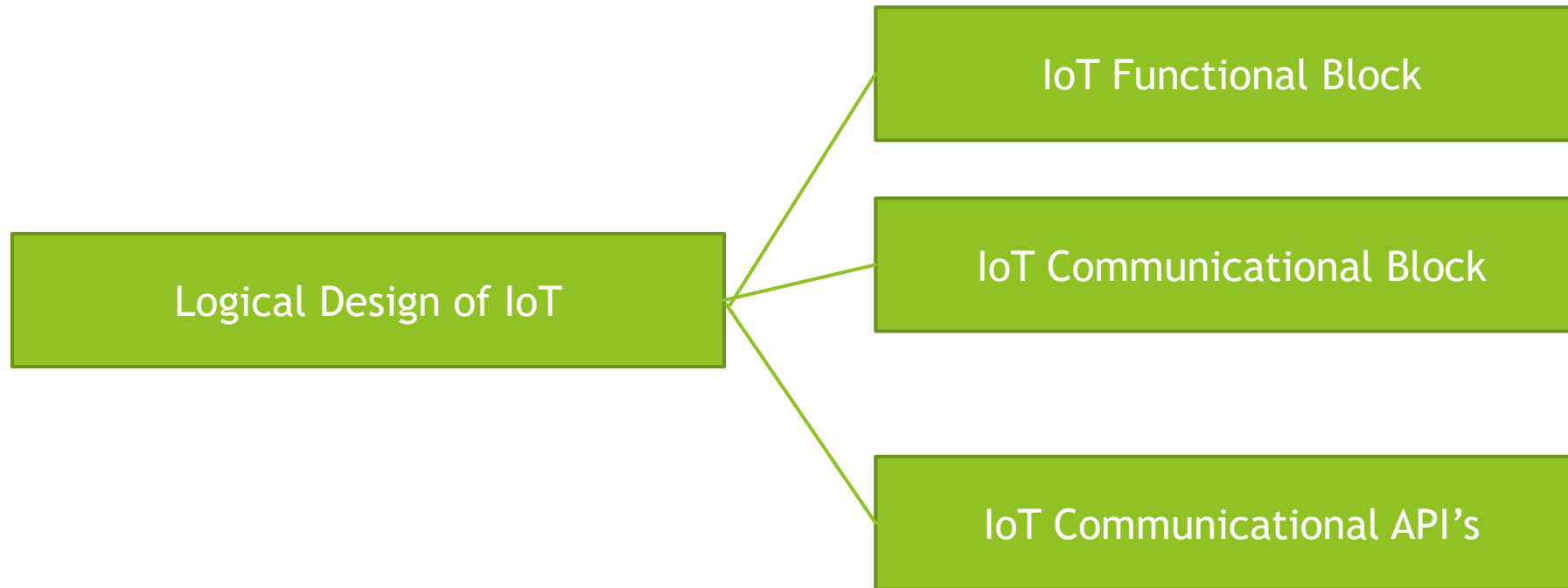


Logical Design of IoT

- ▶ It refers to an abstract representation of the entities and processes without going into the low-level specifics of the implementation.
- ▶ How an IoT system works conceptually..

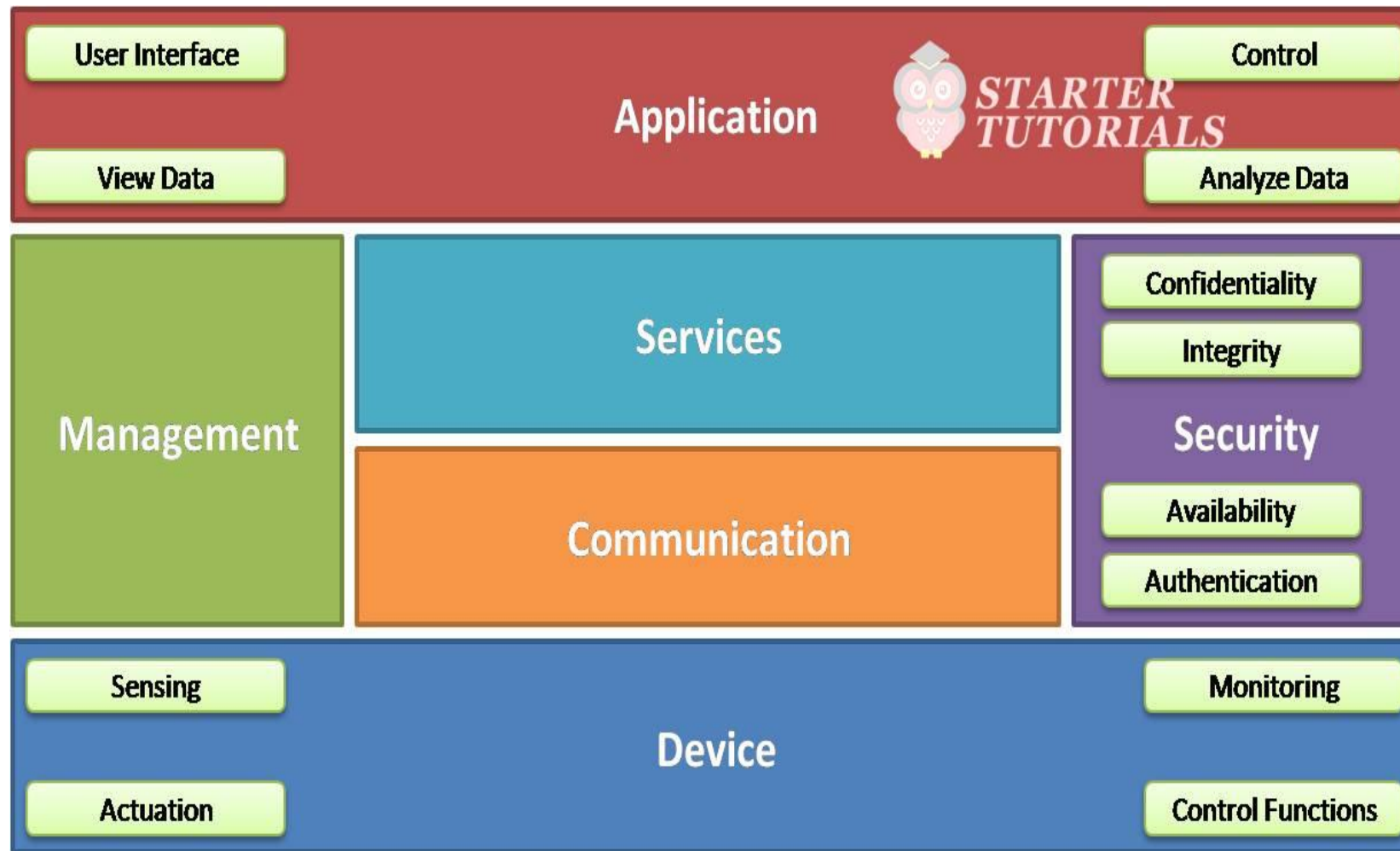


Logical Design of IoT ecosystem



IoT Functional Block

- ▶ An IoT system comprises of a number of functional blocks that provide the system the capabilities for:
 1. Identification
 2. Sensing
 3. Actuation
 4. Communication
 5. Management



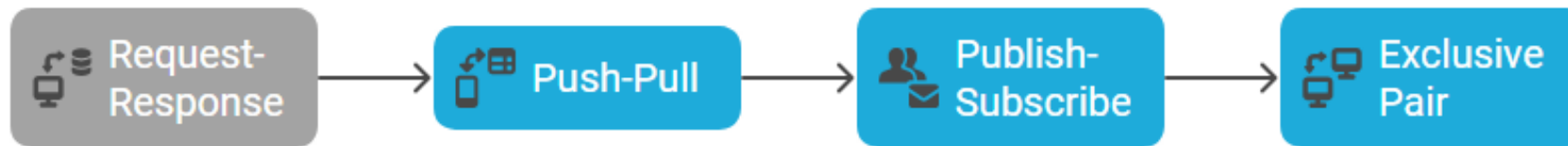
Services

- ▶ Data monitoring services
- ▶ Data control services
- ▶ Data publishing services
- ▶ Device discovery services

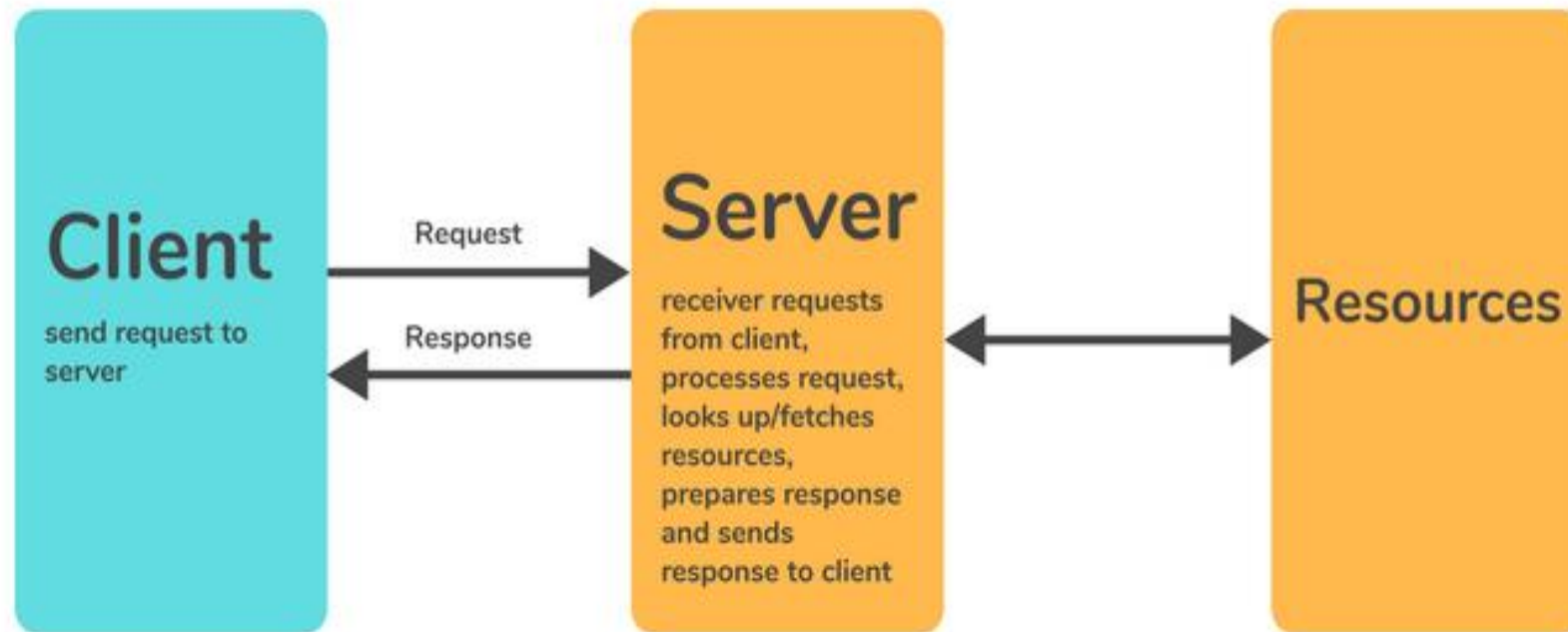
IoT Communicational Block

How devices exchange data?

IoT Communication Models



Request-Response Model



Example

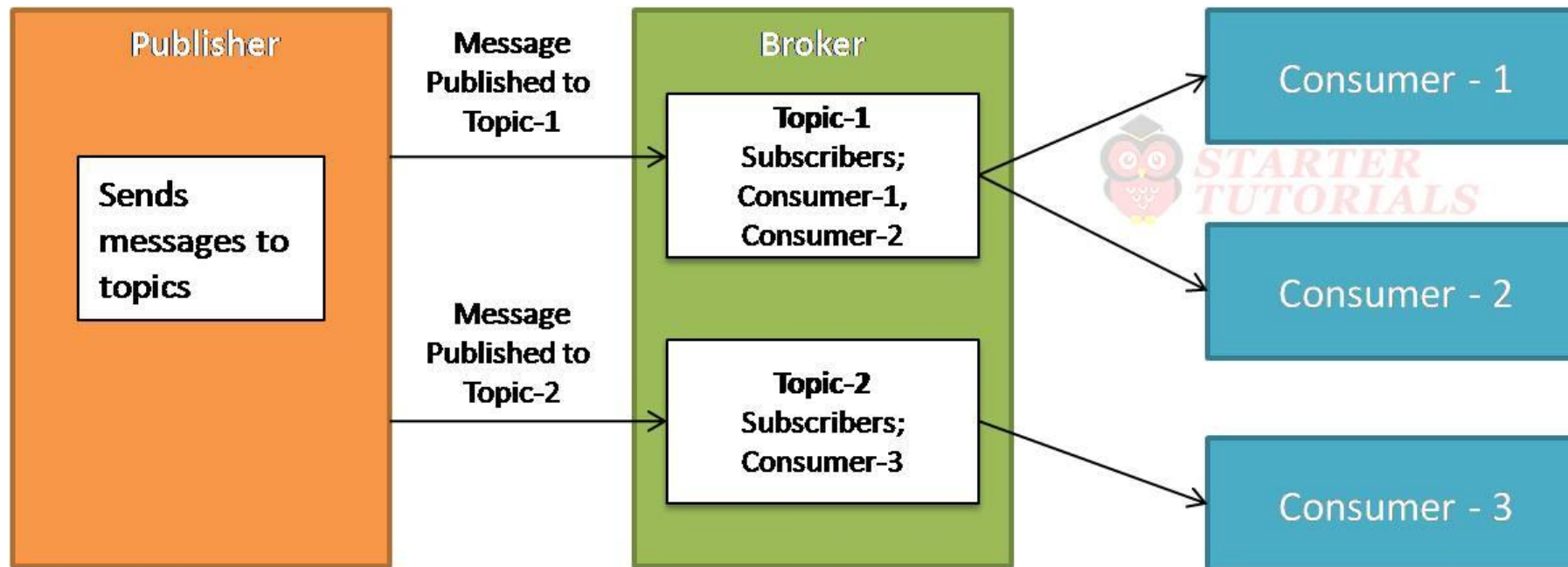
1.

- ▶ Device sends: GET /temperature
- ▶ Server replies: 25° C

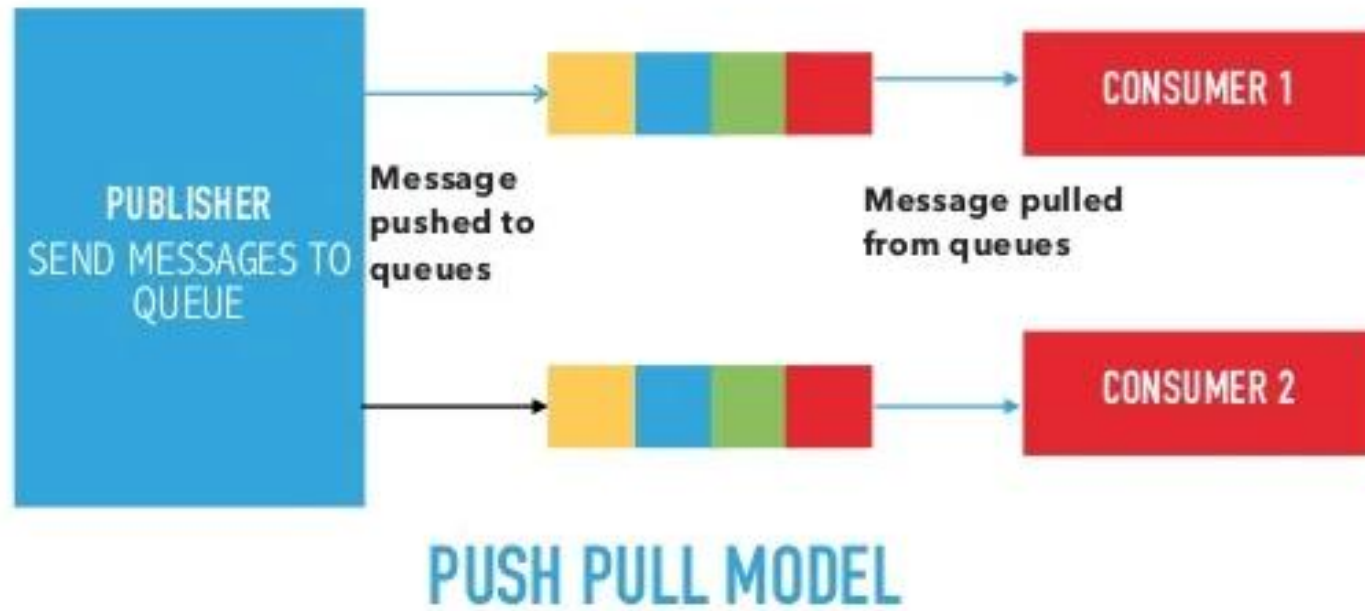
2.

- ▶ App sends request to device: “Turn ON fan”
- ▶ Device responds: “OK, Fan ON”

Publish-Subscribe Model



Push-Pull Model



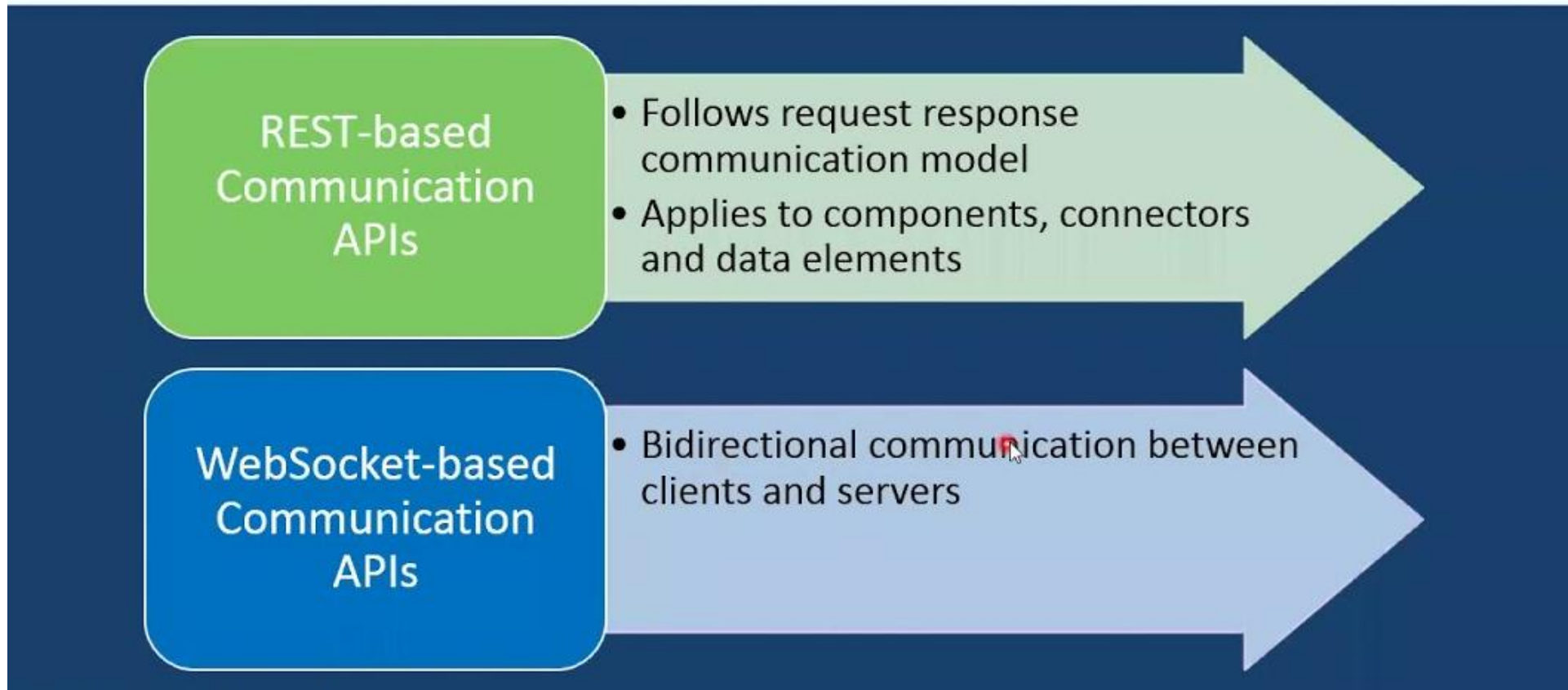
Exclusive Pair Model



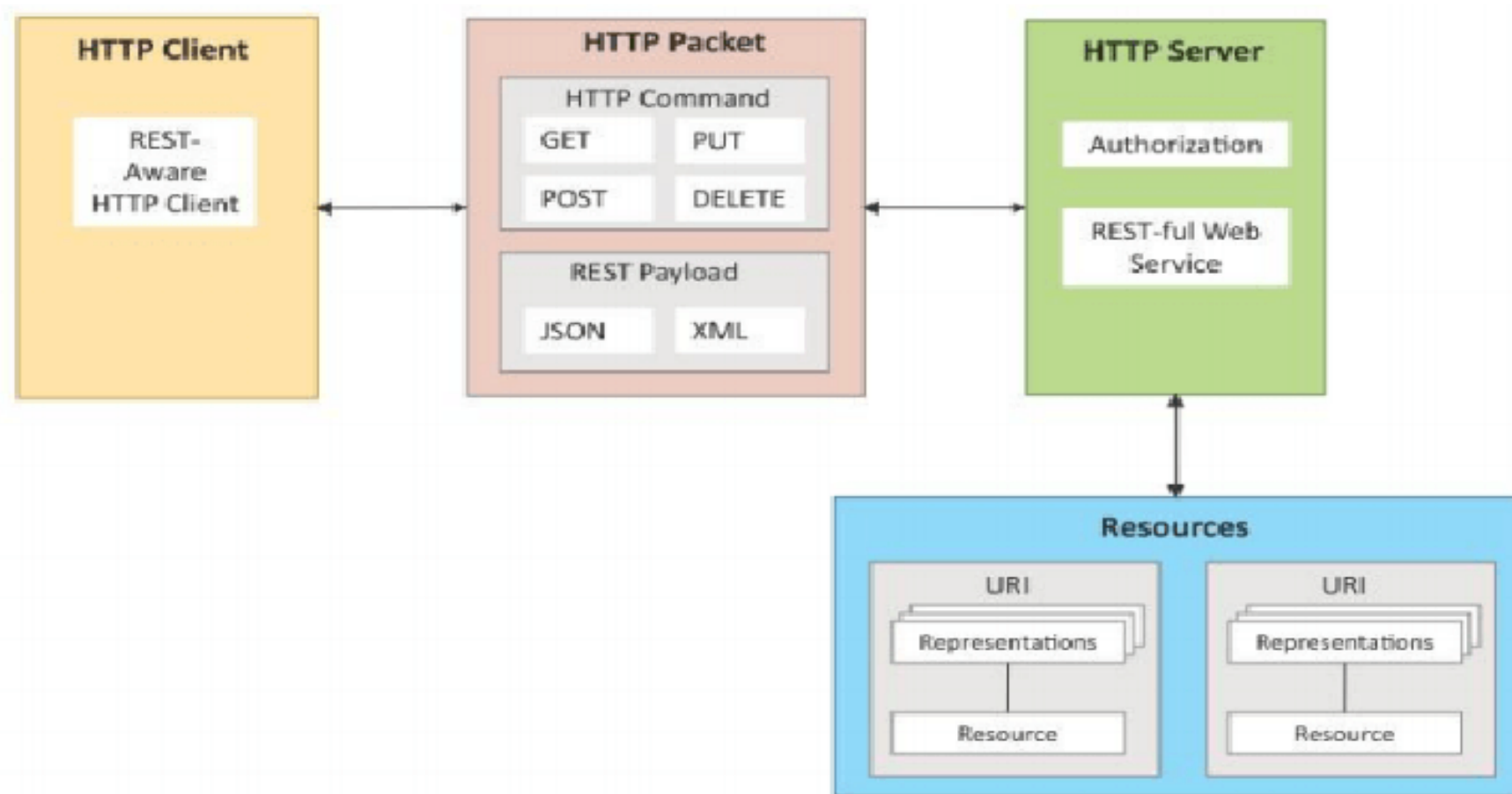
EXCLUSIVE PAIR COMMUNICATION MODEL

IoT Communication APIs

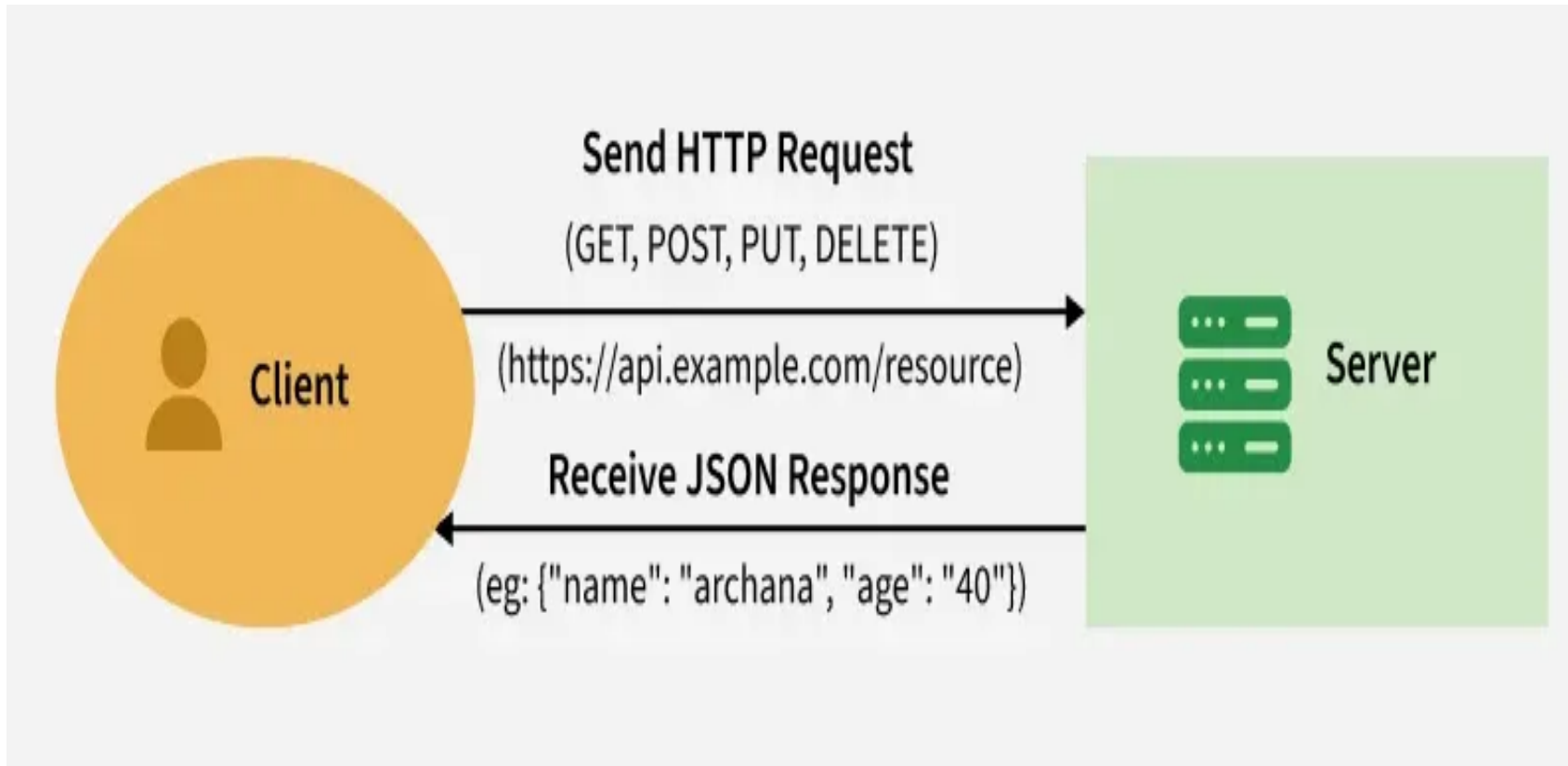
IoT Communication APIs



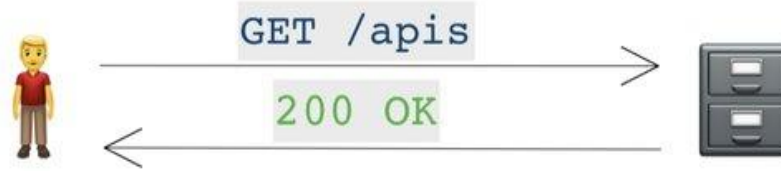
REST (Representational State Transfer)-based Communication APIs



Request-response model used by REST



HTTP Request Methods



GET

The GET method is the most common of all these request methods. It is used to fetch the desired resources from the server.

The POST method is used to submit the information to the server. As we're submitting data, the POST method often changes the state of the server.

POST

PUT

The PUT method is used whenever you need to change the resource. The resource, which is already a part of resource collection.

The PATCH request method is used to modify only the necessary part of the data or response.

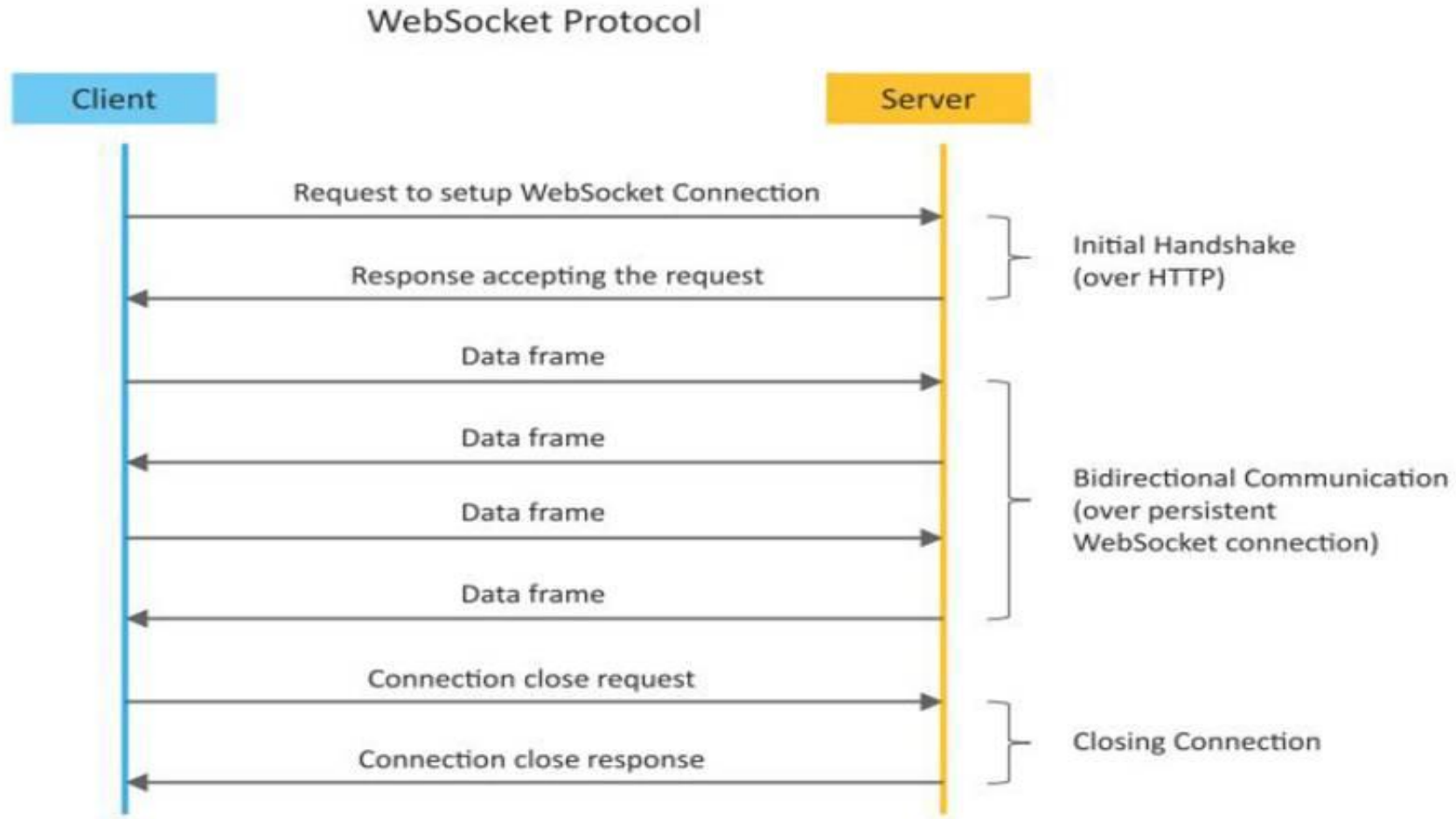
The PATCH method doesn't modify the entire response.

PATCH

DELETE

As the name says, the DELETE request method is used to delete the specified resource. It requests that the origin server delete the resource identified by the Request-URL.

WebSocket-based Communication APIs





Rest API vs WebSocket

Rest API

Request-response

Stateless

CRUD operations, data retrieval

Database interactions, client-server interactions

Communication
Model

State

Use Case

Typical
Applications

WebSocket

Full-duplex, asynchronous

Persistent

Real-time, low-latency
applications

Chat, online gaming, financial
trading, collaborative editing